MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

# Podiatry Diabetic Foot Infection Treatment Workflow

Inclusion Criteria:

- Adult patient (≥ 18 years)
- Infected diabetic foot ulcer
- Patient fits one of the following:
  - Thought to require admission for IV antibiotics
  - Failed PO antibiotics
  - Unable to take PO antibiotics
- Concern for osteomyelitis
- Significantly Immunocompromised: SOT or BMT, pred 20 mg or more per day, neutropenia, 2 or more immunosuppressive agents
- Hemodynamic instability or concern for severe sepsis
- Pregnant




| FQ Dosing | | | | |
|---|---|---|---|---|
| CrCl (mL/min) | Drug | Dose | Frequency | Duration |
| > 50 | Levofloxacin | 750 mg | Daily | 7 days |
| 30-50 | Ciprofloxacin | 750 mg | BID | 7 days |
| 10-29 | Ciprofloxacin | 750 mg | Daily | 7 days |
| < 10 or HD | Ciprofloxacin | 500 mg | Daily | 7 days |
| **Dalbavancin Dosing** | | | | |
| ≥ 30 or HD | Dalbavancin | 1500 mg | x1 | |
| < 30 and not on HD | Dalbavancin | 1125 mg | x1 | |
| **Linezolid dosing** | | | | |
| Any | Linezolid | 600 mg | BID | 7 days |

- Mild infection:
  - lacking any moderate infection features
- Moderate infection:
  - local infection w/erythema > 2 cm or involving deeper structures and no systemic inflammatory response

** See Beta-lactam Allergy Evaluation Service and Empiric Therapy Guidance for patients with allergies

- Risk factors for Gram negative (GN) infection:
  - Previous tissue culture with gram negative bacteria
  - Relapsed or recurrent foot infection
  - Freshwater exposure (lakes, rivers)
  - Recent hospitalization with IV antibiotics (in previous 90 days)

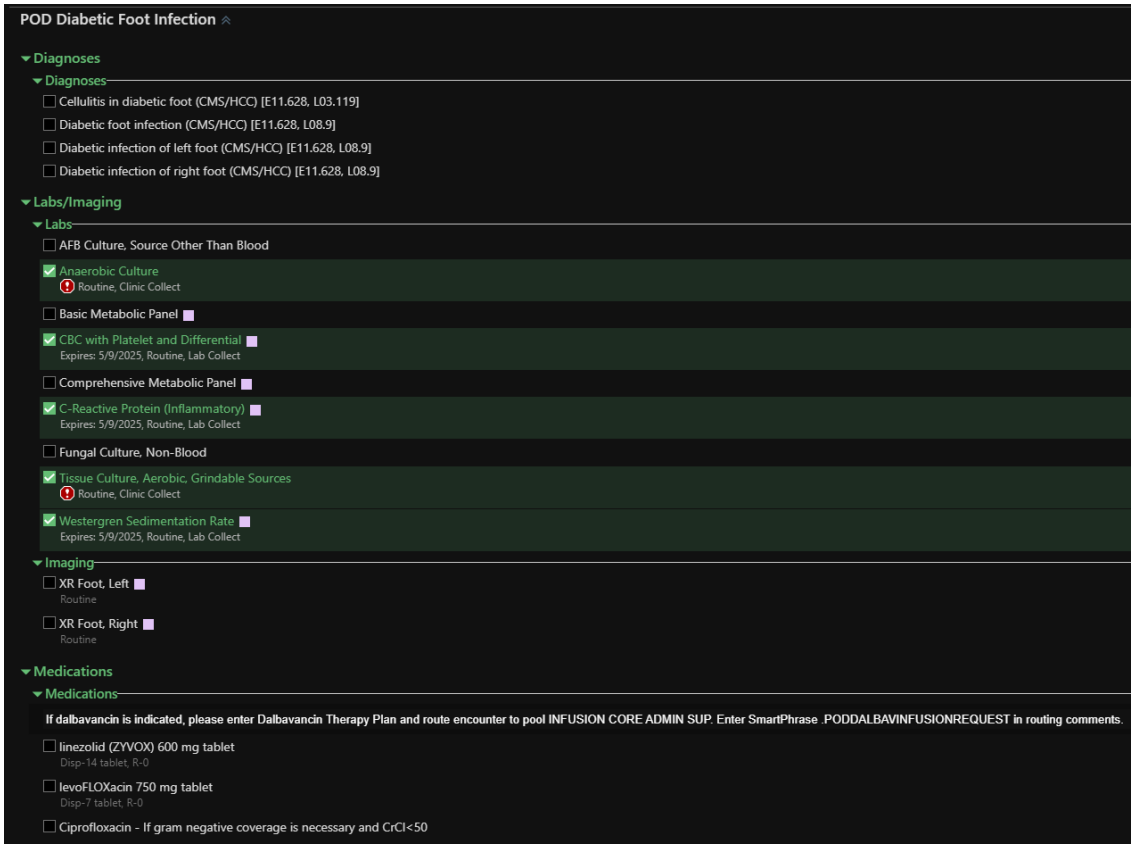
- **Linezolid** availability (consider dalbavancin referral if one of the following):
  - Thrombocytopenia (platelets < 150)
  - Taking MAOi within the past 2 weeks
  - On 2 or more serotonergic agents
    - If on one agent, discuss signs and symptoms of serotonergic syndrome
    - See SSRI & Linezolid Management Guideline

- **Dalbavancin** availability (consider ED referral if one of the following):
  - Weekend visit
  - Weekday afternoon visit (1500 or later M-F)
  - Notified by MIST that patient cannot be scheduled same day
    Allergy to glycopeptide-related medication (vancomycin, telavancin, oritavancin, bleomycin, dalbavancin)
  - Moderate to severe hepatic impairment (Child-Pugh Class B or C)

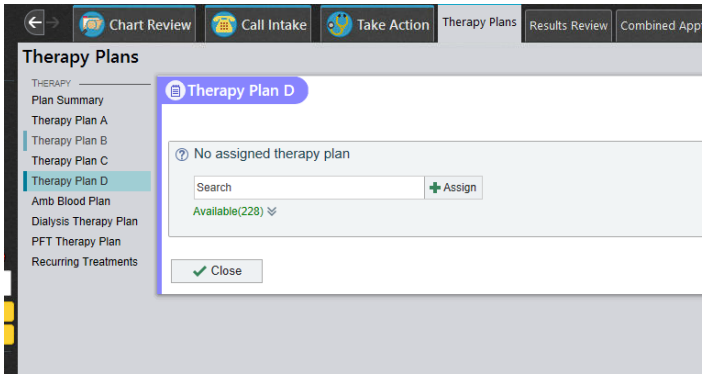
Exclusion criteria:

Process:

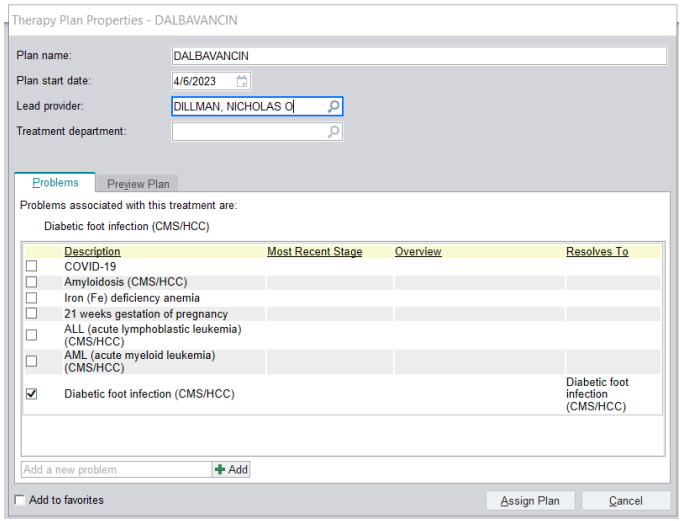
1. Review current/recent tissue cultures if available (previous 6 months)
2. Review inclusion/exclusion criteria above
    a. Any patients who do not meet criteria should not be managed using this protocol
3. Eligible patients can be managed using the 'POD Diabetic Foot Infection' Smartset
4. Upon opening the Smartset, the following sections should be completed:



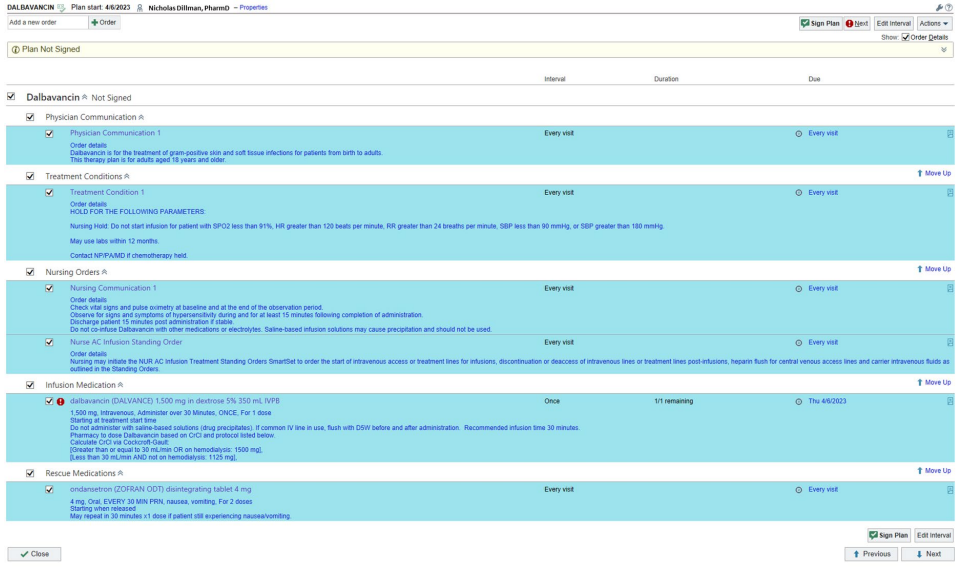

    a. Diagnosis
    b. Labs – standard labs are pre-selected
        i. Consider alternative labs for the following scenarios:
            1. Comprehensive metabolic panel – patients with concern for or known hepatic dysfunction
            2. AFB culture – patients with recurrent DFI
            3. Fungal culture – patients with recurrent DFI

c. Imaging
d. Medications:
    i. Current gram-positive infection only:
        1. Consider **linezolid**
            a. Review exclusions for linezolid:
                i. Thrombocytopenia (platelets < 150)
                ii. Taking MAOi within the past 2 weeks
                iii. On 2 or more serotonergic agents
                    1. If on one agent, discuss signs and symptoms of serotonergic syndrome
                    2. See SSRI & Linezolid Management Guideline for further information on interacting medications and management
            b. If no exclusions for **linezolid**
                i. Order **linezolid** 600 mg PO BID x7 days
            c. If exclusions exist to **linezolid**
                i. Refer to **dalbavancin**
        2. Dalbavancin
            a. Add dalbavancin therapy plan:
                i. Select and open Therapy Plan
                1. 

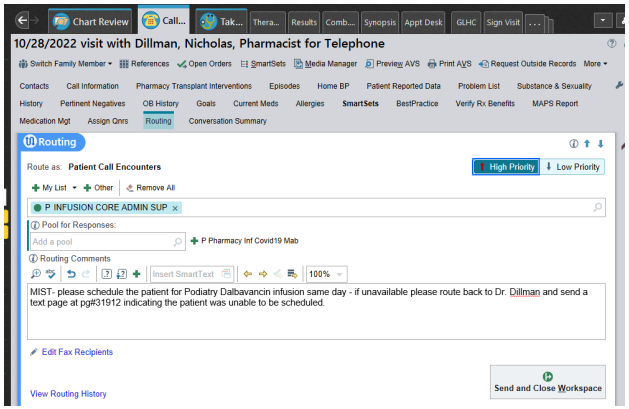
                ii. Search for "**dalbavancin**", input today's date, yourself as lead provider, add 'Diabetic Foot Infection' to problem list if not already listed

1. 


iii. Add indication to **dalbavancin** order, adjust dalbavancin dose for renal function (< 30 mL/min not on dialysis gets a lower dose) and sign order.

1. 


iv. Route to “INFUSION CORE ADMIN SUP” as high priority using “MIST- please schedule the patient for Podiatry Dalbavancin infusion same day – if unavailable please route back to Dr. *** and send a text page at pg#*** indicating the patient was unable to be scheduled.” in the message section.

1. 


   b. If **dalbavancin** is unavailable (after 1300 M-F, weekends, or infusion availability is limited):
      i. Refer to ED for further care

ii. Current gram-negative infection only:
   1. Review susceptibilities, if FQ susceptible:
      a. Review CrCl and prescribe **FQ** as follows:

| CrCl (mL/min) | Drug | Dose | Frequency | Duration |
|---|---|---|---|---|
| > 50 | **Levofloxacin** | 750 mg | Daily | 7 days |
| 30-50 | **Ciprofloxacin** | 750 mg | BID | 7 days |
| 10-29 | **Ciprofloxacin** | 750 mg | Daily | 7 days |
| < 10 or HD | **Ciprofloxacin** | 500 mg | Daily | 7 days |

   2. If not **FQ** susceptible
      a. Refer to ED

iii. Mixed gram-positive and gram-negative infection OR no current cultures
   1. Follow gram positive pathway under Process 4.d.i.
   2. Review risk factors for gram negative infection:
      a. Relapsed or recurrent foot infection
      b. Freshwater exposure (lakes, rivers)
      c. Recent hospitalization with IV antibiotics (within previous 90 days)
   3. If gram negative risk factors present, follow gram negative only pathway Process 4.d.ii.

5. Print and provide patient education document with standard supportive care/patient education
   a. Specifically, highlight the importance of leg elevation and rest.
6. Patient should be scheduled for follow-up visit with Podiatry in 2-3 days

| **Last revised:** 04/2023 | **Antimicrobial Stewardship Subcommittee:** N/A | **P&T:** N/A |
|---|---|---|

| **References** | |
|---|---|
| | **Citation** |
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| **Revision History** | |
|---|---|
| **Date** | **Revision** |
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*The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.*